Israel Souza Ferreira

Data Scientist | Machine Learning Engineer

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Professional Summary

Computer Science graduate with experience in applied Machine Learning research, computer vision, and systems development. I compare baselines and run experiments, calibration, automated testing, and preparation for CPU deployment.

Education

Bachelor of Science in Computer Science

Federal University of Ceará (UFC) — Quixadá Campus, Brazil

Dates: 2021 – August 2026. Completed on August 19, 2026.

Undergraduate thesis: Metaheuristics and Integer Linear Programming for Triple Roman Domination in Graphs; developed in 2024–2025 and defended in February 2025.

Technical Skills

Languages and data: Python, SQL, C++, Bash, Pandas, NumPy, PostgreSQL, pgvector.

Machine Learning and computer vision: PyTorch, scikit-learn, SciPy, OpenCV, YuNet, SFace, ONNX Runtime, OCR, embeddings, cross-validation, calibration, and ablation.

Engineering: FastAPI, SQLAlchemy, Alembic, Docker, GitHub Actions, Git, pytest.

Professional Experience

Research and Development (R&D) Researcher

Tieta.ai — Remote, Brazil

July 2026 – Present

Research Fellow in the Research and Innovation Training Program.

- Contributed to research software development and maintenance; implemented validation and automated tests.
- Automated technical activities and produced versioned documentation to support reproducibility and traceability.

Full Stack Developer and Machine Learning Engineer

Independent consulting for an educational institution — Remote, Brazil

March 2026 – Present

Project: Face Clock Evoluir. Contracted, functional system in final acceptance testing.

- Built models, FastAPI APIs, PostgreSQL/pgvector persistence, security, tests, and the deployment package for facial attendance and medical-document triage. The delivery record contains 574 automated tests.
- Calibrated YuNet/SFace 1:N facial recognition: accepted unknowns decreased from 45.4% to 0.2% on public data. CPU benchmark p95 was 74 ms with one simultaneous identification and 1.87 s with 20; real-population calibration is still required.
- Compared five medical-document classifiers on 64 test documents. EfficientNet with Logistic Regression achieved F1 of 0.778 and accuracy of 87.5%; false positives fell from 9 to 2 versus the original neural head, maintaining 70% recall (14/20 medical documents).
- Reduced total delivery images from 10.59 to 1.11 GB (about 89.5% less disk) by matching dependencies to a GPU-less server. ONNX Runtime migration preserved classifier decisions on the compared test set.

Machine Learning Researcher
Tieta.ai — Remote, Brazil
April 2025 – June 2026
CNPq RHAE Research Fellow, part-time.

- Conducted applied Machine Learning research, developing and evaluating computational solutions.
- Ran experiments and analyzed results; established reproducible development and evaluation processes.

Teaching Assistant — Algorithm Design and Analysis
Federal University of Ceará — Quixadá Campus — Quixadá, Brazil
March 7, 2023 - November 30, 2023

- Supported undergraduate students in algorithms and complexity analysis.

Teaching Assistant — Calculus
Federal University of Ceará — Quixadá Campus — Quixadá, Brazil
April 2022 – July 2022

- Supported undergraduate students with Calculus concepts and exercises.

Selected Projects

Triple Roman Domination in Graphs — Undergraduate Thesis | 2024–2025

- Implemented genetic algorithm (GA) and ant colony optimization (ACO) metaheuristics in C++, with a Python/Pyomo integer linear programming model. Proposed correcting a published formulation that accepted invalid solutions; evaluated 362 graphs with one run per instance.

Code: https://github.com/isrreal/Triple-Roman-Domination-in-graphs

Brazilian Emergency Aid — Data Engineering | In development

- Implemented Pandas and asyncpg COPY chunked ingestion for a 31.6 GB public source. On subsets of up to 5 million rows, measured post-cleanup RSS growth from about 182 to 750 MiB, using fixed 100-thousand-row chunks and three measured runs.
- Observed an association between RSS and retained IDs (r approximately 0.991); correlation does not establish causality. PostgreSQL staging-based deduplication is the next experiment, with no published improvement yet.

Code: https://github.com/isrreal/AuxilioEmergencialQueries

Languages

Portuguese: native. English: good communication; fluent technical reading of research papers and documentation.